EBU6018

Some revision of first half of teaching plan.

Answer the following questions, including mathematical definitions where appropriate.

1. List the various forms of the Fourier Transform.

2. What is the use of the Fourier Transform?

3. What is the principal limitation of the Fourier Transform?

4. What are the basis functions of the Fourier transform and why are they a good choice of basis functions?

5. What is meant by the term “Windowed Transform”?

6. What advantage is gained by using a windowed transform?

7. What is the implication of the Uncertainty Principle?

    i) In general
    ii) Specifically related to windowed transforms.

8. Compare the mathematical expressions for the Fourier Transform, the Short Time Fourier Transform and the Wavelet Transform.

9. What is a Wavelet?

10. What advantage does a wavelet transform have over a Short Time Fourier Transform?

11. Explain the difference between a Gabor Transform and a Morlet Wavelet Transform. What is the advantage of a Morlet Wavelet Transform?

11. How can a Wavelet Transform be used for “Feature Extraction”?

12. How can a Wavelet Transform be used for “Trend Analysis”?

13. How can a Wavelet Transform be used for Compression?

14. What is a Spectrogram?

15. What is a Scalogram?

16. What is the relationship between the time/frequency axes of the Spectrogram and the translation/scale axes of the Scalogram?

17. What is “Multiresolution Analysis (MRA)”?

18. What is a “Filter Bank” and how can it be used to implement Wavelet Transforms?

19. What are Haar Functions? Use a Haar Matrix to explain how they can be used for Feature Extraction.

20. What information does a Discrete Cosine Transform provide?